ECOVISION AI: Environmental Intelligence & Abatement Report

Department of Computer Science & Engineering (AIML) & Environmental Engineering

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Report Type: Automated Analytical Assessment

Dataset Records: 2,009

Standard: CPCB National Ambient Air Quality Standards (NAAQS)

1. Executive Summary & Problem Statement

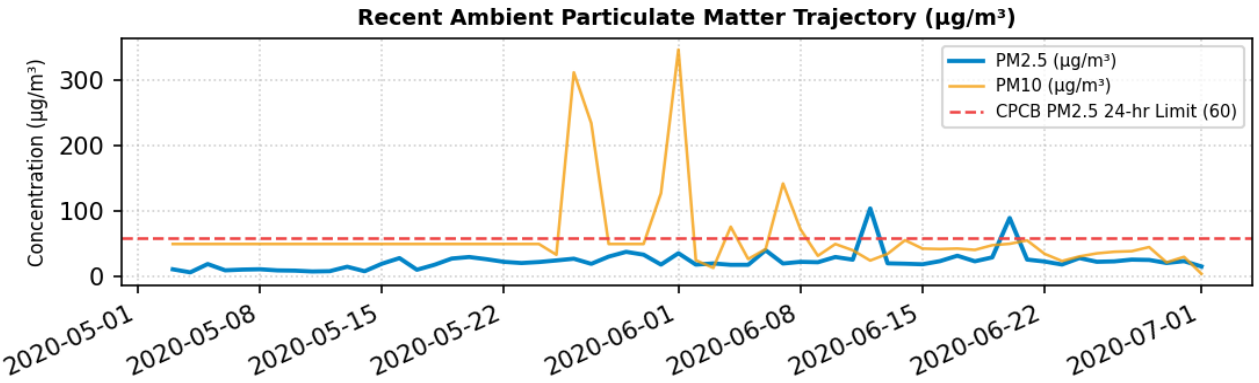
EcoVision AI provides an automated, end-to-end framework for ambient air pollution monitoring, predictive modeling, statistical anomaly isolation, and explainable engineering abatement synthesis. Urban industrial centers experience severe public health consequences due to fine particulate (PM2.5) exposure. This platform bridges environmental data science with industrial abatement technologies by converting raw sensor telemetry into verifiable mitigation actions.

2. Environmental Dataset & Data Dictionary

Variable	Type	Unit	CPCB 24-hr Benchmark	Environmental Significance
PM2.5	Float	µg/m³	60.0 µg/m³	Fine inhalable combustion aerosol
PM10	Float	µg/m³	100.0 µg/m³	Fugitive road dust & mechanical aggregates
NO2	Float	µg/m³	80.0 µg/m³	Thermal combustion from vehicle & power plant stacks
SO2	Float	µg/m³	80.0 µg/m³	Sulphur oxidation from coal & heavy fuel oils
CO	Float	mg/m³	2.0 mg/m³	Incomplete hydrocarbon combustion marker
O3	Float	µg/m³	100.0 µg/m³ (8-hr)	Secondary photochemical smog oxidant

3. Observed Pollutant Statistics

Pollutant	Observed Mean	CPCB Limit	Compliance Status
PM2.5	50.14 µg/m³	60.0 µg/m³	**Compliant**
PM10	52.13 µg/m³	100.0 µg/m³	**Compliant**
NO2	16.83 µg/m³	80.0 µg/m³	**Compliant**
SO2	7.85 µg/m³	80.0 µg/m³	**Compliant**
CO	1.06 mg/m³	2.0 mg/m³	**Compliant**
O3	32.4 µg/m³	100.0 µg/m³	**Compliant**



4. Predictive Machine Learning Models (PM2.5 Estimation)

Model Architecture	MAE (µg/m³)	RMSE (µg/m³)	R² Coefficient	Generalization Summary
Linear Regression (Baseline OLS)	15.51	18.78	0.1382	Linear baseline; limited capturing non-linear interactions
Random Forest Regressor (Ensemble)	13.09	17.07	0.2878	Superior capture of multi-pollutant non-linear dependencies

5. Anomaly Detection Analysis (Scikit-learn Isolation Forest)

Evaluated 2,009 telemetry records. Isolated **61** multivariate statistical outliers (3.04% of dataset).

Date	Location / Station	PM2.5	PM10	NO2	Isolation Score	Primary Sensor Spikes
2015-04-28	Chennai	70.85	50.24	45.19	-0.1834	NO2 (45.19 µg/m³), SO2 (32.85 µg/m³), CO (13.1 µg/m³)
2015-03-27	Chennai	266.29	50.24	50.12	-0.1417	PM2.5 (266.29 µg/m³), NO2 (50.12 µg/m³), SO2 (25.45 µg/m³)
2016-09-20	Chennai	22.2	50.24	34.72	-0.1294	NO2 (34.72 µg/m³), CO (11.03 µg/m³)
2015-03-06	Chennai	45.54	50.24	16.34	-0.1174	CO (11.02 µg/m³), O3 (105.13 µg/m³)

6. AI Abatement Action Recommendations

Recommendation 1: Industrial High-Efficiency Particulate Filtration (Fabric Filters & ESP)

Triggering Pollutant: PM2.5 | **Observed:** 50.14 µg/m³ (Standard: 60.0 µg/m³) | **Risk:** Moderate (Within Threshold)

Scientific Rationale: Fine particles (aerodynamic diameter <= 2.5 µm) penetrate deeply into alveolar tissue. Pulse-jet baghouses and high-voltage Electrostatic Precipitators (ESP) trap sub-micron particulates via inertial impaction, Brownian diffusion, and electrostatic charging with >99.5% efficiency.

Expected Environmental Gain: Reduces ambient PM2.5 loading by 35-60% in immediate industrial downwind zones.

Recommendation 2: Combustion Source Clean Transition & Stubble/Biomass Burning Controls

Triggering Pollutant: PM2.5 | **Observed:** 50.14 µg/m³ (Standard: 60.0 µg/m³) | **Risk:** Moderate (Within Threshold)

Scientific Rationale: Combustion of solid fuels (coal, biomass, pet-coke) produces dense aerosols of black carbon and polycyclic organic matter. Transitioning to piped natural gas (PNG) or electric induction prevents primary carbonaceous aerosol formation.

Expected Environmental Gain: Lowers localized combustion-derived aerosol concentrations by 40-70%.

7. Methodological Limitations & Conclusion

Limitations & Scientific Scope: Telemetry imputation utilizes localized medians and temporal interpolation. Estimated PM2.5 concentrations are statistical models based on ground-level precursor observations and do not model synoptic meteorological dispersion, solar radiation flux, or thermal inversion layer height. **Conclusion:** Continuous real-time telemetry processing coupled with explainable machine learning provides municipal and industrial environmental managers with verifiable insights for preemptive pollution abatement.

8. Regulatory & Academic References

1. Central Pollution Control Board (CPCB), Ministry of Environment, Forest and Climate Change, Govt. of India, *National Air Quality Index (NAQI) Standard Report*, 2014.
2. World Health Organization (WHO), *WHO Global Air Quality Guidelines: Particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide*, Geneva, 2021.
3. Breiman, L., 'Random Forests', *Machine Learning*, 45(1), 5-32, 2001.
4. Liu, F. T., Ting, K. M., & Zhou, Z. H., 'Isolation Forest', *IEEE ICDM*, 413-422, 2008.